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Dynamics of Targeted Misinformation Propagation in Closed Homogeneous Minority Communities with Small World Networks

Emergent Research Forum (ERF) Paper

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Abstract

This paper investigates the nuanced dynamics of misinformation spreading in social networks, with a specific focus on the propagation of targeted misinformation within a closed homogeneous minority community characterized by a small world network. In contrast to epidemic spreading, misinformation can be tailor-made to target specific groups, amplifying its impact. The study introduces an I2S2R model, distinguishing between targeted and non-targeted information propagation structures. To emulate a community structure, the model considers a high clustering coefficient. The basic reproduction number is considered the outcome variable to assess the influence of the community structure on misinformation propagation. Additionally, the model explores the interplay between two types of misinformation over a specified period. This research contributes to understanding the complex relationship between community structures and targeted misinformation dynamics, offering an important overview of the potential vulnerabilities and impacts within minority communities.

Keywords

Misinformation spreading, targeted misinformation, minority, community, SIR, closed homogeneous network.

Introduction

How does misinformation spread within minority communities compared to non-minority groups? This crucial question lies at the center of our investigation, guiding us to explore the deeper understanding of misinformation dissemination in small ethnic minority groups utilizing a basic Susceptible-Infectious-Recovered (SIR) model and manipulating the clustering coefficient. Additionally, we delve into whether minority communities show a higher susceptibility to targeted misinformation compared to their non-targeted counterparts. Furthermore, we address the interrelation between targeted and non-targeted misinformation within minority communities, employing a novel S2I2R model. This exploration is crucial for unraveling the complexities of misinformation dynamics and paving the way for informed interventions within minority contexts.

Extensive prior research has investigated various aspects of information diffusion, but this investigation concentrates primarily on minority communities. While minority communities may vary in specific characteristics such as language, socioeconomic status, or historical experiences, they are all united by the common experience of being marginalized or treated differentially compared to the dominant group in society (Britannica, 2024). This study is important because it sheds light on the spread of false information within ethnic minority groups. Fake news that targets these communities raises concerns

about potential defamation of groups and harm to dignitaries, issues that are being recognized and addressed more and more in different parts of the world. This study aims to contribute to existing literature by guiding targeted interventions and suggesting policy measures aimed at reducing the detrimental effects of misinformation on minority communities.

In this study, we consider fake news, misinformation, and rumors as interchangeable terms, all representing deceptive information aimed at misleading people. While traditional epidemiological models assume homogeneity, recent research acknowledges the significance of community structure in understanding misinformation dynamics, as highlighted by Zhang et al. (2014), which has also been the approach followed by this study. In this context, targeted misinformation refers to fraudulent information deliberately tailored to influence or manipulate members of minority communities, leveraging their distinctive cultural, linguistic, or socioeconomic qualities to spread false narratives.

Literature Review and Hypothesis Development

Classic Epidemic Models in Epidemiology: The groundwork on modeling infectious diseases comes from Kermack and McKendrick (1927), who introduced two classic epidemiology models: the SIS (Susceptible-Infectious-Susceptible) and the SIR (Susceptible-Infectious-Recovered) models. These models categorize the population into three groups: Susceptible (S), Infectious (I), and Recovered (R). Over the past decade, the SIR model has been successfully applied to model information diffusion on social networks (Lerman and Ghosh 2010).

Homogenous Small World Network: A social network is called a *small-world network* if, roughly speaking, any two people in the network can reach each other through a short sequence of acquaintances (Kleinberg 2004). Milgram's (1967) basic small-world experiment is the most famous experiment that analyzed the small-world problem. Santos et al. (2005) first introduced a class of small-world networks—homogeneous small-worlds—which, in contrast with the well-known Watts-Strogatz (1998) small-worlds, exhibit a homogeneous connectivity distribution in the sense that all nodes have the same number of connections. This feature allows the investigation of pure small-world effects, detached from any associated heterogeneity. The homogeneity assumption means that the probabilities of state transitions are uniform across all individuals. In our study, we assume the minority community to have a small world homogeneous network where the clustering coefficient is high, and the degree of node is fixed among all nodes.

Epidemic Models for Information Diffusion: Among the extensive literature on information diffusion, this study deliberately concentrates on minority communities. Given the extensively documented human inclination to accept false information in social settings, especially when it originates from "in-group sources," as outlined by Flynn et al. (2017). Again, minority communities that often feel marginalized by mainstream representation tend to retreat into more private digital spaces, like social media groups, where misinformation spreads unchecked (Fernando, 2023). Therefore, we hypothesize:

Hypothesis 1: Misinformation is disseminated more within minority communities compared to non-minority communities.

Individuals tend to categorize themselves and others into social groups based on shared characteristics, such as ethnicity or race. Minority communities often develop a strong sense of social identity due to shared experiences of marginalization or discrimination. Targeted misinformation that aligns with the narratives or beliefs of these minority groups can serve to reinforce their sense of identity and solidarity. As a result, members of minority communities may be more likely to share and propagate misinformation that resonates with their group identity, leading to the faster dissemination of targeted misinformation within these communities compared to non-targeted misinformation.

Hypothesis 2: Targeted misinformation is disseminated more within minority communities compared to non-targeted misinformation.

Existing models typically focus on single rumors, with exceptions like Wang et al. (2014), who introduced a 2SI2R model involving the simultaneous spread of two rumors. Their findings suggest that Rumor2 inhibits Rumor1, and a stronger Rumor2 force accelerates Rumor1's decline. Conversely, when Rumor1 is negative and Rumor2 is its contrary, Rumor1's spread strengthens under increased Rumor2 force. Ran and Chen (2021) proposed a PaNSEIR model considering positive and negative information interference,

but none addressed targeted misinformation toward a special group. This study, the first to incorporate two messages simultaneously, meaning the group is exposed to both misinformation at the same time—one targeted and the other general—hypothesizes that targeted misinformation facilitates non-targeted misinformation, building on observed effects in previous studies.

Hypotheses 3: Targeted misinformation facilitates the dissemination of non-targeted misinformation within minority communities.

Model 1 (Basic SIR)	Model 2(S2I2R)
$\frac{dS}{dt} = -\beta \cdot S \cdot I \cdot P$ $\frac{dI}{dt} = \beta \cdot S \cdot I \cdot P - \gamma I$ $\frac{dR}{dt} = \gamma I$	$\frac{dS}{dt} = -\beta_T(1 - P_1) \cdot S \cdot I_T - \beta_{NT} \cdot (1 - P_2) \cdot S \cdot I_{NT}$ $\frac{dI_T}{dt} = \beta_T \cdot P_1 \cdot S \cdot I_T + \delta I_T I_{NT} \cdot Q + \lambda I_T R_{NT} U - \gamma_T I_T R_T$ $\frac{dI_{NT}}{dt} = \beta_{NT} \cdot P_2 \cdot S \cdot I_{NT} - \delta I_T I_{NT} \cdot Q - \gamma_{NT} R_{NT} \cdot I_{NT} (1 - Q)$ $\frac{dR_T}{dt} = \gamma_T I_T R_T + \beta_T (1 - P_1) \cdot S$ $\frac{dR_{NT}}{dt} = \gamma_{NT} R_{NT} \cdot I_{NT} (1 - Q) - \lambda I_T R_{NT} U + \beta_{NT} (1 - P_1) \cdot S$
where, $S+I+R=N=1$ $-\beta \cdot S \cdot I \cdot P$ represents the rate at which susceptible individuals transition to the infected state (I) due to exposure to misinformation. γI represents the rate at which informed individuals recover or fact-check the misinformation. This term accounts for individuals who become less influenced by misinformation over time. β represents the transmission rate and γ represents the recovery rate.	$\frac{dS}{dt}$ represents the rate of change of the susceptible compartment. $\frac{dI_T}{dt}$ & $\frac{dI_{NT}}{dt}$ represents the rate of change of the infectious compartment for targeted and non-targeted misinformation. $\frac{dR_T}{dt}$ & $\frac{dR_{NT}}{dt}$ represents the rate of change of the recovered compartment for targeted and non-targeted misinformation respectively. β_T, β_{NT} is transmission rate, γ_T & γ_{NT} are the recovery rate, P_1 and P_2 are the probability of a susceptible to be infected respectively for targeted and non-targeted misinformation. Q is the probability of an infected person by non-targeted misinformation to be infected by targeted misinformation. U is the probability of a recovered person from non-targeted misinformation being infected by targeted misinformation.

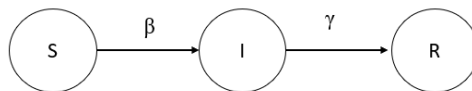


Figure 1. Model 1: Basic S-I-R model

Research Model & Analysis

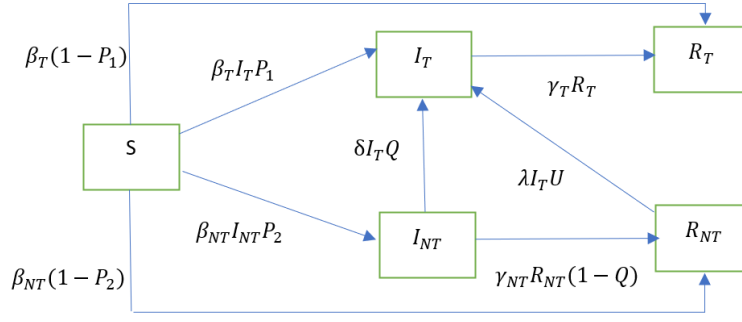


Figure 2. S2I2R model

Underlying Assumptions and Analysis Parameters

Susceptibility

All individuals within the minority community are considered initially susceptible to both targeted and non-targeted misinformation. It is assumed that the minority community is more susceptible to targeted misinformation. Consequently, individuals initially infected by non-targeted misinformation have a probability of subsequent infection by targeted misinformation. The study acknowledges the potential for complex interactions between the two types of misinformation. Individuals recovering from non-targeted misinformation are considered susceptible to subsequent infection by targeted misinformation. However, the reverse scenario, where individuals recovering from targeted misinformation are infected by non-targeted misinformation, is not explicitly modeled, assuming a minimal immunity threshold against non-targeted misinformation.

Network

The study operates within the framework of a closed homogeneous network, where the total number of nodes is constant and there are no external influences such as births or deaths. This assumption simplifies the modeling approach and focuses on the internal dynamics of the minority community. The population is assumed to exhibit homogeneous mixing, implying that every individual has an equal probability of interacting with any other individual in the community. This assumption facilitates a simplified representation of the social network structure.

Basic Reproduction Number

The basic reproduction number R_0 is a critical epidemiological metric used to evaluate the potential spread of infectious diseases or, in this context, misinformation within a population. It represents the average number of secondary infections generated by one infected individual in a completely susceptible population. In our models, R_0 serves as a key indicator of the potential for misinformation dissemination. If $R_0 > 1$, it suggests that each case of misinformation leads to more than one additional case, indicating the potential for an epidemic or widespread dissemination. In our first model, R_0 is calculated as the ratio of the base transmission rate (β) to the recovery rate (γ). For the second model, which involves targeted misinformation, calculating R_0 is more complex and requires determining the dominant eigenvalue $\rho(F)$ of the next-generation matrix.

From that can be calculated using the formula: $R_0 = \frac{\rho(F)}{\gamma_T R_T + \lambda R_{NT} U}$. We introduce the clustering coefficient (C), which represents the tendency of nodes in a network to cluster together and may influence transmission rates. In our simulations for the first research question, we vary β based on the clustering coefficient (C) to understand its impact on misinformation spread in the first model using the relation: $\beta = \beta_0 \cdot C$

Conclusion

This study makes several significant contributions to the understanding of misinformation dynamics within minority communities. Firstly, by introducing and applying an innovative S2I2R model, we provide a novel framework for examining the simultaneous spread of targeted and non-targeted misinformation. This model serves as a valuable tool for capturing the complexities of misinformation propagation within specific social contexts, shedding light on differential susceptibility and the interplay between misinformation types. Secondly, our research lays the groundwork for future simulations and real-world experiments by employing the SIR and S2I2R models. These models, combined with synthetic network data, offer a versatile platform for exploring misinformation dynamics under various scenarios, facilitating insights into mitigation strategies and intervention planning. Additionally, the findings generated from simulations can provide a basis for designing controlled experiments within online community groups of ethnic minorities, enabling empirical validation and refinement of the proposed models. The practical implications of the findings include informing targeted interventions and policy recommendations to mitigate the impact of misinformation on minority populations. Future research opportunities involve expanding the model's applicability, incorporating real-world validation data, and considering ethical and cultural nuances in further investigations. Overall, this study contributes to advancing our understanding of misinformation in digital communication and underscores the importance of context-aware research in addressing societal challenges.

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